

(a) Encoded biological functions and mechanisms

Detect	Checkpoint	Repair / recover	Terminal response
Animal / human			
ATM / ATR damage channels	CHK1/2 p53-p21	HR/NHEJ BER/NER	Apoptosis

Arabidopsis

ATM / ATR damage channels	SOG1-WEE1 SMR checkpoint	HR/NHEJ Excision repair	SMR induction then differentiation / endoreduplication*
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*One combined terminal transition in the plant encoding, not two alternatives.

(b) Same structures, different observational resolution

Readout	What is distinguished	PN-GDDA	Branching relation
A Coarse	Shared functions; terminal loss of proliferative capacity	0.9976	Weakly bisimilar
B Pathway	Damage, ATM/ATR and repair; terminal mechanisms collapsed	0.9976	Weakly bisimilar
C Terminal	Pathway events plus separate animal / plant terminal mechanisms	0.9976	Non-comparable (both directions fail)

Model- and interface-conditional results; no organism-level equivalence